

Ygeiopolis Healthcare System - Improvements To Do

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Team planning document for the final database submission.

The core schema is now strong: constraints, triggers, stored procedures, validation checks, portable setup scripts, and a larger data generator are in place. The remaining work is mostly final-submission packaging, query writing, evidence gathering, and report polishing.

1. Highest Priority Before Submission

Priority

Task

Why It Matters

Expected Output

P0

Create the final SQL query files for Q01 to Q15.

The assignment grades the query section heavily. Each query must return valid rows, otherwise it will not be graded.

sql/Q01.sql through sql/Q15.sql .

P0

Run every query against a freshly loaded database and save output files.

The deliverable explicitly asks for one output text file per query.

sql/Q01_out.txt through sql/Q15_out.txt .

P0

Prepare the Q4 and Q6 optimization evidence.

Q4 and Q6 require EXPLAIN or EXPLAIN ANALYZE, an alternative FORCE INDEX or hint version, cost/runtime comparison, and a short conclusion.

Report section with screenshots or copied plans, plus the two SQL variants.

P0

Confirm the official reference data story.

The brief says ICD-10, KEN, drugs, and medical procedure reference data should come from official sources, not invented names/codes.

A short table in the report listing official, cleaned, fallback, and synthetic data.

P0

Create final diagrams in the exact required format.

The final zip requires ER and relational diagrams as PDFs.

diagrams/er.pdf and diagrams/relational.pdf .

2. Query Work Plan

Suggested split: one person owns correctness for Q1-Q5, one person Q6-Q10, one person Q11-Q15. Then everybody reviews the full set together.

Query Group

Main Tables/Views

Notes

Q1 revenue by department/year/KEN/insurance

hospitalization , department , ken , patient

Split base KEN cost from extra charge caused by stay length above MDN.

Q2, Q5, Q11 doctor/procedure workload
doctor , personnel , doctor_department , shift_assignment , procedure_event
Use current-year filtering consistently. Explain any date assumption in README/report.

Q3, Q6, Q9 patient history and stay length
hospitalization , patient , department , icd10_diagnosis , hospitalization_evaluation
Use TIMESTAMPDIFF or equivalent day calculation consistently.

Q4 doctor evaluations
hospitalization_doctor , hospitalization_evaluation , doctor , personnel
Must include optimization comparison because Q4 has the special EXPLAIN requirement.

Q7 and Q10 drug/allergy/substance analysis
active_substance , patient_allergy , drug_active_substance , prescription , v_prescription_substances
For Q10, build pairs of active substances prescribed during the same hospitalization.

Q8 and Q12 shift coverage
v_shift_roster , v_shift_coverage , department_shift , staff subtype tables
Use the views to keep the final SQL readable.

Q13 supervision hierarchy
doctor , personnel
Use a recursive CTE to walk supervisor chains from direct supervisor to director .

Q14 ICD-10 categories over consecutive years
hospitalization , icd10_diagnosis
Group by the first three ICD-10 characters and compare adjacent years.

Q15 triage distribution
emergency_visit , department
Calculate waiting time, hospitalization percentage, and referral department distribution.

3. Data And Reproducibility Tasks

Run the generator with the final selected size and keep the produced bundle stable until submission.

Use the portable command `mysql --local-infile=1 -u root -p < sql/setup.sql` from the bundle root.

Verify that `sql/validation.sql` returns zero rows for problem-detection queries.

Save `dataset_summary.json` and mention the row counts in the report.

If the EMA Article 57 file is not used, clearly mark the drug dataset as demo fallback in the assumptions/report.

4. Report And Presentation Tasks

Task
Recommended Owner
Output

Write schema design explanation and assumptions.
Database design owner

Report sections: ER model, relational model, assumptions.

Write SQL implementation explanation.

SQL implementation owner

Constraints, triggers, procedures, views, indexes.

Write data-generation explanation.

Data owner

Official sources, cleaning, synthetic data, reproducibility.

Prepare screenshots and query outputs.

Query owners

Q01-Q15 result evidence and Q4/Q6 EXPLAIN evidence.

Final proofread and zip structure check.

Whole team

Final package matching the assignment structure.

5. Nice-To-Have Improvements

Add small negative trigger tests that intentionally fail inside transactions and then roll back.

Add EXPLAIN notes for all important queries, not only Q4 and Q6.

Prune any indexes that are not used by final queries after the query set is complete.

Add final diagrams/ folder with clean PDF diagrams following the required names.

Create a short demonstration script with the exact commands used during grading.

6. Suggested Final Order Of Work

Generate and load final data.

Run validation and fix any returned problem rows.

Write Q01-Q15.

Save Q01-Q15 output files.

Run Q4/Q6 EXPLAIN and FORCE INDEX variants.

Finish diagrams and report.

Check final folder names against the assignment PDF.